

# Business Requirements Document (BRD)

Checkout Process Optimization Project (ShopEase)

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## **1. Document Overview**

### **1.1 Purpose**

This document defines the business requirements for improving the checkout process of ShopEase, an e-commerce platform experiencing high cart abandonment rates. The goal is to reduce friction in the checkout journey and improve conversion rates using data-driven insights.

### **1.2 Background**

Data analysis from user sessions (Python, SQL, Power BI) revealed that approximately 70% of users abandon the checkout process. Major drop-offs occur at the payment stage, with additional friction caused by mandatory login requirements and multi-step checkout flow.

### **1.3 Business Objective**

- Reduce cart abandonment rate by 20%
- Improve checkout conversion rate
- Enhance mobile user experience
- Simplify payment and checkout process

## **2. Scope**

### **2.1 In Scope**

- Checkout process redesign
- Guest checkout implementation
- One-page checkout flow
- Payment process optimization
- UI/UX improvements for mobile users

## **2.2 Out of Scope**

- Product catalog redesign
- Pricing strategy changes
- Marketing campaigns
- Inventory management system

## **3. Stakeholders**

- Product Manager
- Marketing Manager
- IT/Development Team Lead
- Customer Support Head
- Finance Manager

## **4. Current Problem Summary (Data Insights)**

- 49% cart abandonment rate
- Highest drop-off at payment stage
- Mobile users show highest abandonment
- Mandatory login creates early friction
- Multi-step checkout reduces conversion rate

## 5. Functional Requirements

| ID   | Requirement Description                                                         |
|------|---------------------------------------------------------------------------------|
| FR1  | System shall allow users to checkout as guests without mandatory login          |
| FR2  | System shall provide a one-page checkout process combining shipping and payment |
| FR3  | System shall support multiple payment methods (card, wallet, etc.)              |
| FR4  | System shall display a progress indicator during checkout                       |
| FR5  | System shall allow users to save payment details securely                       |
| FR6  | System shall optimize checkout UI for mobile devices                            |
| FR7  | System shall reduce number of form fields in checkout process                   |
| FR8  | System shall auto-fill user details for returning customers                     |
| FR9  | System shall validate payment details in real-time                              |
| FR10 | System shall send confirmation after successful checkout                        |
| FR11 | System shall allow users to edit cart during checkout                           |
| FR12 | System shall handle payment failure with retry options                          |

## 6. Non-Functional Requirements

- System shall load checkout pages within 3 seconds
- System shall ensure 99.9% payment system uptime
- System shall follow PCI-DSS security compliance
- System shall be mobile responsive
- System shall encrypt all payment data

7. Assumptions

- Users have internet access and valid payment methods
- Payment gateway is reliable and integrated
- Users prefer faster checkout experience

Requirements Traceability Matrix (RTM)

| Requirement ID | Description              | Source (Data Insight / Business Need)     | Priority |
|----------------|--------------------------|-------------------------------------------|----------|
| FR1            | Guest checkout           | High abandonment due to login requirement | High     |
| FR2            | One-page checkout        | Funnel drop-off at multiple steps         | High     |
| FR3            | Multiple payment methods | Payment stage abandonment                 | High     |
| FR4            | Progress indicator       | User confusion in checkout steps          | Medium   |
| FR5            | Save payment details     | Improve return customer conversion        | Medium   |
| FR6            | Mobile optimization      | High mobile abandonment rate              | High     |
| FR7            | Reduce form fields       | UX friction identified in analysis        | High     |
| FR8            | Auto-fill details        | Reduce checkout time                      | Medium   |
| FR9            | Real-time validation     | Payment errors causing drop-off           | High     |
| FR10           | Order confirmation       | Standard business requirement             | High     |
| FR11           | Cart editing             | Improve flexibility during checkout       | Medium   |
| FR12           | Payment retry option     | Reduce abandonment from payment failure   | High     |

## 9. Conclusion

This BRD defines a structured solution to reduce cart abandonment and improve checkout efficiency. The Requirements Traceability Matrix ensures that every functional requirement is directly linked to real user behavior insights, making the solution data-driven and business-focused.